

## **AKTU Lucknow**

### **SUPPLEMENTARY EXAMINATION ? 2025**

Machine Learning ? Supplementary Examination, 2025

Subject Code: CS502 Semester: 5 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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#### **SECTION A (Unit 1: Supervised Learning)**

1. Derive the expression related to Supervised Learning and explain each term. [12 Marks]
2. List the advantages and limitations of Supervised Learning. Explain any two in detail. [12 Marks]
3. Compare and contrast the key concepts involved in Supervised Learning. [12 Marks]

#### **SECTION B (Unit 2: Unsupervised Learning)**

4. Describe the working principle of Unsupervised Learning with necessary equations. [12 Marks]
5. Write short notes on Unsupervised Learning. [12 Marks]
6. Derive the expression related to Unsupervised Learning and explain each term. [12 Marks]

#### **SECTION C (Unit 3: Model Evaluation and Tuning)**

7. Describe the working principle of Model Evaluation and Tuning with necessary equations. [12 Marks]
8. Solve a problem related to Model Evaluation and Tuning. Show all the steps clearly. [12 Marks]
9. Compare and contrast the key concepts involved in Model Evaluation and Tuning. [12 Marks]

#### **SECTION D (Unit 4: Ensemble Methods)**

10. Explain Ensemble Methods with a neat diagram/sketch and a suitable example. [8 Marks]
11. Describe the working principle of Ensemble Methods with necessary equations. [12 Marks]
12. How is Ensemble Methods implemented? Give an algorithm / procedure and explain. [8 Marks]

#### **SECTION E (Unit 5: Neural Network Basics)**

13. Write short notes on Neural Network Basics. [12 Marks]
14. Define Neural Network Basics. Explain its significance in Machine Learning with a

**suitable example. [12 Marks]**

15. Define Neural Network Basics. Explain its significance in Machine Learning with a suitable example. [12 Marks]

\*\*\* End of Question Paper \*\*\*